Title: Hybrid Optimization for Supply Chain Inventory Management Using Deep Reinforcement Learning and Multi-Stage Stochastic Programming

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Abstract

Supply Chain Inventory Management (SCIM) involves making sequential decisions under uncertainty to balance production, storage, and logistics costs while satisfying fluctuating customer demand. Existing approaches, such as multi-stage stochastic programming (MSP) and deep reinforcement learning (DRL), each offer unique strengths but suffer from scalability and adaptability limitations when applied independently. This report presents a novel hybrid framework that integrates DRL for long-term production planning with MSP for short-term logistics optimization. We formulate a two-echelon supply chain problem and implement a simulation environment reflecting real-world complexities, including seasonal demand, production constraints, and storage limitations. Extensive numerical experiments are conducted under basic and advanced settings, using optimality gap and EVPI-gap as evaluation metrics. Results demonstrate that our proposed DRL-MSP method outperforms standalone baselines in both cost efficiency and robustness, particularly in scenarios with high uncertainty. This work offers an effective solution for complex SCIM tasks and contributes a reproducible benchmark for future research.

1. Introduction

Supply Chain Inventory Management (SCIM) is a fundamental problem in operations research and logistics, requiring the coordination of production, storage, and transportation under uncertainty. The objective is to minimize total operational costs while ensuring timely fulfillment of customer demand. SCIM becomes particularly challenging when faced with stochastic and seasonal demand patterns, multi-echelon structures, capacity constraints, and complex logistics decisions.

Traditional solutions to SCIM primarily include Multi-Stage Stochastic Programming (MSP) and Reinforcement Learning (RL). MSP formulates the problem as a scenario-based optimization model that explicitly accounts for uncertainty (Khouja, 2003; Preusser et al., 2010). Although MSP provides strong theoretical guarantees, it suffers from scalability issues due to the exponential growth of the scenario tree with the number of stages and variables. On the other hand, RL methods have been employed to learn adaptive decision-making strategies from interactions with the environment, but classical RL often encounters convergence instability, high sensitivity to hyperparameters, and poor generalizability in real-world SCIM settings (Hubbs et al., 2020).

Recent advances in Deep Reinforcement Learning (DRL) have improved the scalability and expressiveness of RL models, enabling their application in complex, high-dimensional decision environments (Gijsbrechts et al., 2022). However, existing DRL approaches in SCIM remain limited in their ability to handle key practical factors such as seasonal demand, production and storage capacity constraints, and variable logistics costs. Moreover, many DRL implementations struggle to incorporate domain-specific constraints in a structured and interpretable way.

This report investigates whether a hybrid framework that combines DRL and MSP can offer a scalable, adaptive, and cost-effective solution to SCIM under realistic operational conditions. To this end, we propose a novel heuristic that uses DRL to learn long-term production decisions and MSP to solve short-term logistics subproblems under uncertainty. This structure allows us to exploit the adaptability of DRL and the optimization strengths of MSP in a complementary manner. The approach is evaluated in a simulated two-echelon supply chain environment, comprising a central factory and multiple distribution warehouses, as illustrated in Figure 1 (Preusser et al., 2010).

The goal of this study is threefold: first, to design a DRL-MSP hybrid framework tailored for inventory decision-making in seasonal and stochastic supply chain environments; second, to implement a simulation environment that reflects realistic constraints such as demand variability, production and storage limits, and logistic costs; and third, to evaluate the proposed method through numerical experiments, comparing it against classical policies and standalone DRL baselines using metrics such as total cost, optimality gap, and EVPI-gap.

The results show that the proposed hybrid method outperforms traditional approaches, especially in complex scenarios where purely MSP- or DRL-based methods become less effective. Overall, this research demonstrates that a hybrid methodology not only enhances performance and scalability but also offers better interpretability in supply chain inventory optimization under uncertainty.

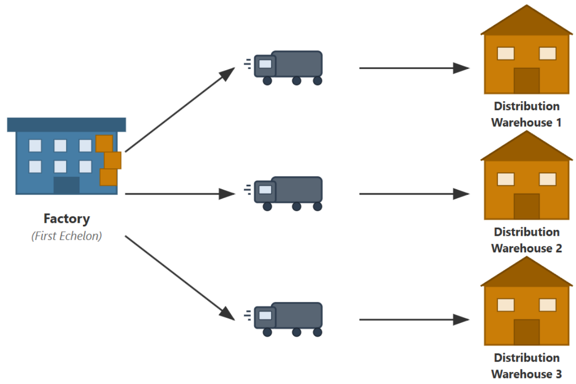

Figure 1. A two-echelon supply chain consisting of a factory (first echelon) and three distribution warehouses (second echelon) (Preusser et al., 2010).

Table 1. Parameters defining the SCIM environment. The two values correspond to experiments with $J = 5$ and $J = 10$ warehouses, respectively.

Parameters	Setting
Maximum demand value	2
Production costs	1
Vehicles capacities	2
Logistic costs fixed	random(0.7, 1)
Logistic costs variable	random(0.01, 0.07)
Maximum production	{13, 25}
Factory capacities	{18, 36}
Distribution warehouses capacities	{6, 6}
Factory storage costs	random(0.01, 0.1)
Distribution warehouses storage costs	random(1, 2)
Backorder costs	10

2. Data Set and Task

2.1. Data Set

As real-world datasets and benchmarks are not readily accessible, we generate synthetic yet realistic data to create a controlled environment for experimentation. In our research, we employ the SCIMAI-Gym environment as our primary experimental dataset (Hubbs et al., 2020; Gijsbrechts et al., 2022). We selected this environment for its comprehensive representation of real-world supply chain dynamics and its suitability for reinforcement learning applications (Peng et al., 2019; Kemmer et al., 2018). Our implementation builds upon the framework introduced by Stranieri et al. (Stranieri et al., 2024), which effectively combines deep reinforcement learning with multi-stage stochastic programming.

To simulate realistic operational scenarios, we evaluate algorithm performance across two settings of complexity. The *basic setting* considers two distribution warehouses ($J = 2$) over a time horizon of seven timesteps (days), while the *advanced setting* involves five or ten distribution warehouses ($J = 5$ or $J = 10$) over twelve timesteps (months), modeling a realistic annual supply cycle. Each configuration is run for 250 independent episodes to ensure statistical significance. Key environment parameters are summarised in Table 1. These values are chosen to reflect real-world constraints while preserving proportional scaling between the $J = 5$ and $J = 10$ scenarios. This allows for a fair comparison across different supply chain sizes and complexities.

For demand generation, we implement a seasonal and stochastic model that combines sinusoidal demand patterns with random variation (Preusser et al., 2010; Khouja, 2003). The amplitude parameter D exceeds maximum production capacity, ensuring capacity constraints are active. The phase parameter is fixed at zero to synchronize peak demand across warehouses, representing a challenging allocation scenario.

2.2. Task

We model a single-item, two-echelon supply chain under periodic review, consisting of a central factory and a set of distribution warehouses:

$$\mathcal{J} = \{1, 2, \ldots, J\},$$

where decisions include how much to produce and how to distribute it at each timestep $t \in \{0, 1, \ldots, T-1\}$. The system evolves based on production, inventory, logistics, and uncertain demand (Alonso-Ayuso et al., 2003; Zipkin, 2000).

The total cost at each timestep is given by:

$$\begin{aligned} C_t = \underbrace{c_0\, x_t}_{\text{production cost}} + \underbrace{\sum_{j=1}^{J} l_j\, z_{j,t}}_{\text{variable logistic cost}} + \underbrace{\sum_{j=1}^{J} p_j\, y_{j,t}}_{\text{fixed logistic cost}} \\ + \underbrace{\sum_{j=0}^{J} h_j \max[I_{j,t},\, 0]}_{\text{storage cost}} - \underbrace{\sum_{j=0}^{J} b_j \min[I_{j,t},\, 0]}_{\text{backorder cost}} \end{aligned} \tag{1}$$

Inventory is updated by:

$$\begin{aligned} I_{0,t+1} &= \min(I_{0,t} + x_t,\, I_0^{\max}) - \sum_{j=1}^{J} z_{j,t}, \\ I_{j,t+1} &= \min(I_{j,t} + z_{j,t-1},\, I_j^{\max}) - d_{j,t}, \quad \forall j = 1, \ldots, J. \end{aligned} \tag{2}$$

All notation used in these equations is summarized in Table 2. The environment's state space includes inventory levels at all locations and three steps of historical demand. Actions include production quantity x_t and distribution amounts $z_{j,t}$, constrained by storage and capacity limits (Peng et al., 2019).

To evaluate policy performance, we use:

- **Total cost:** Average cost over episodes.

Table 2. Notation Table

Symbol	Description	Units / Notes
t	Discrete time index (timestep)	$t = 1, 2, \ldots, T$
T	Total number of timesteps in one planning horizon	Integer
J	Number of distribution warehouses	Integer
x_t	Number of batches produced at time t	Batches
c_0	Production cost per batch	Cost/batch
$z_{j,t}$	Number of batches shipped from the factory to warehouse j at time t	Batches
l_j	Variable logistic cost per batch shipped to warehouse j	Cost/batch
$y_{j,t}$	Number of vehicles used to ship batches to warehouse j at time t	Vehicles
p_j	Fixed logistic cost per vehicle used for warehouse j	Cost/vehicle
$I_{j,t}$	Inventory level at the factory ($j = 0$) or at warehouse j at time t	Batches (can be negative)
h_j	Storage cost per batch at location j	Cost/batch/time
b_j	Backorder (penalty) cost per unit of unmet demand at location j	Cost/batch
$d_{j,t}$	Demand at warehouse j during time t	Batches
$I_j^{\max}$	Maximum storage capacity at location j	Batches
C_t	Total cost incurred at time t	Cost
V_j	Vehicle capacity for shipments sent to warehouse j	Batches/vehicle
min, max	Standard operators indicating the minimum or maximum value of a set	—

- **Opt-gap:** Used in the basic scenario:

$$\text{Opt-gap} = \frac{\text{Cost}_{\text{alg}} - \text{Cost}_{\text{MSP}}}{\text{Cost}_{\text{MSP}}} \times 100\% \tag{3}$$

- **EVPI-gap:** Used in the advanced scenario:

$$\text{EVPI-gap} = \frac{\text{Cost}_{\text{alg}} - \text{Cost}_{\text{PI}}}{\text{Cost}_{\text{PI}}} \times 100\% \tag{4}$$

These metrics (Hubbs et al., 2020) quantify how closely an algorithm approaches optimal or ideal performance given the scenario complexity.

3. Methodology

This project employs a hybrid approach combining Deep Reinforcement Learning (DRL) and Multi-Stage Stochastic Programming (MSP) to optimize supply chain inventory management (SCIM). The goal is to minimize total costs while ensuring efficient stock replenishment across a two-echelon divergent supply chain, as discussed in multi-stage, multi-customer frameworks (Khouja, 2003), and illustrated in Figure 2.

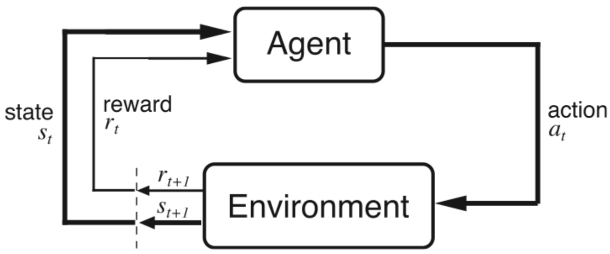

Figure 2. Agent-environment interface (Sutton et al., 1998).

3.1. Deep Reinforcement Learning (DRL)

SCIM is formulated as a Markov Decision Process (MDP), consisting of a state, an action, and a reward:

State: The state at timestep t is defined as

$$s_t = (I_{0,t}, I_{1,t}, \ldots, I_{J,t}, d_{t-1}, \ldots, d_{t-\tau}), \tag{5}$$

where $I_{j,t}$ represents the inventory at warehouse j, d_t denotes past demand values, and τ is the time window length.

Action: The DRL agent selects production and shipment decisions:

$$a_t = (x_t, z_{1,t}, \ldots, z_{J,t}). \tag{6}$$

Reward: The negative total cost,

$$R_t = -C_t, \tag{7}$$

where C_t includes production, storage, transportation, and backorder costs.

As an initial exploration, we consider the Asynchronous Advantage Actor–Critic (A3C) algorithm, which employs multiple worker processes that each interact with the environment in parallel. These workers continuously update a global network, thereby stabilizing the learning process and providing diverse experiences. A3C's asynchronous framework helps speed up training and is particularly beneficial in large or high-dimensional environments. However, in practice, A3C can exhibit higher variance in policy updates and may require careful hyperparameter tuning to achieve convergence.

While A3C provides a solid baseline for policy-gradient methods, we employ Proximal Policy Optimization (PPO) as our main DRL approach. PPO optimizes the objective function

$$L^{\text{CLIP}}(\theta) = \mathbb{E}_t\left[\min\Big(r_t(\theta)\,A_t,\ \text{clip}(r_t(\theta),\,1-\epsilon,\,1+\epsilon)\,A_t\Big)\right], \tag{8}$$

where $r_t(\theta) = \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{\text{old}}}(a_t|s_t)}$ is the importance sampling ratio, and A_t is the advantage function. The clipping function prevents overly large policy updates, improving training stability.

The training process begins with the DRL agent interacting with the environment to collect transition data in the form of (s_t, a_t, R_t, s_{t+1}). Based on this collected data, the policy π_θ is updated using the Proximal Policy Optimization (PPO) algorithm, aiming to maximize the expected cumulative reward while maintaining training stability through clipped policy updates. The refined policy is then used to generate new production decisions x_t, which are subsequently fed into the MSP module to determine optimal short-term transportation and inventory actions under demand uncertainty.

3.2. Multi-Stage Stochastic Programming (MSP)

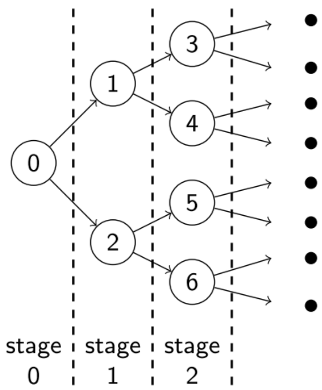

Figure 3. Scenario tree representation used in MSP. Each node corresponds to a possible demand realization at a given stage, enabling stage-wise adaptive decision-making under uncertainty (Stranieri et al., 2024).

In our framework, MSP is employed to optimize short-term logistics decisions by explicitly modeling demand uncertainty via a scenario tree (Preusser et al., 2010). As illustrated in Figure 3, each node in the tree represents a potential realization of demand at a specific decision stage. The branches represent stochastic transitions, and the tree structure evolves over a discrete time horizon. This formulation enables stage-wise and adaptive decision-making, where each decision must be made based on current and past information, adhering to the non-anticipativity principle.

By capturing future uncertainties through a finite set of demand scenarios, MSP allows the system to preemptively plan logistics actions that are robust across multiple possible futures. This is particularly important in supply chains with volatile or seasonal demand patterns, where short-term decisions—such as how much to ship or how many trucks to dispatch—must be made without full knowledge of future outcomes.

The objective of MSP is to minimize the expected total cost across all scenarios:

$$\min \mathbb{E}\left[c_0 x + \sum_{j=1}^{J}\left(l_j z_j + p_j y_j + h_j I_j - b_j B_j\right)\right], \quad (9)$$

where c_0 is the unit production cost, l_j is the per-unit shipping cost to warehouse j, p_j is the fixed cost per vehicle used, h_j is the per-unit storage cost, and b_j is the penalty cost for each unit of backordered demand B_j.

This optimization is subject to several operational constraints:

- **Inventory balance constraint**:

$$I_j = I_j^{\text{prev}} + z_j - d_j, \quad \forall j, \quad (10)$$

  ensuring that inventory at each warehouse reflects incoming shipments and outgoing demand.

- **Capacity constraints**:

$$0 \le x \le X_{\text{max}}, \quad 0 \le z_j \le I_{\text{max},j}, \quad \forall j, \quad (11)$$

  which impose upper bounds on production and storage capacity.

- **Vehicle constraints**:

$$z_j \le V_j y_j, \quad \forall j, \quad (12)$$

  which couple the number of batches shipped z_j with the number of vehicles y_j and their respective capacities V_j.

Here, x denotes the production quantity at the factory, z_j is the number of batches shipped to warehouse j, y_j is the number of trucks or transportation units used, and I_j is the updated inventory level at location j after satisfying demand d_j. Backordered demand B_j may occur when demand exceeds the available inventory, resulting in penalty costs.

By solving this MSP model at each time step with updated production inputs from the DRL agent, the framework ensures that logistics decisions—shipping quantities, vehicle assignments, and inventory updates—are optimized in a way that is both cost-effective and resilient to demand fluctuations. This integration of forward-looking stochastic optimization with real-time demand scenarios makes MSP a powerful component for tactical planning in supply chain systems.

3.3. Integration of DRL and MSP

The proposed hybrid approach integrates deep reinforcement learning (DRL) for long-term production planning with multi-stage stochastic programming (MSP) for short-term logistics optimization, achieving both flexibility and robustness in inventory management (Khouja, 2003). DRL determines production quantities x_t based on system observations such as inventory levels and recent demand patterns.

Among available DRL algorithms, we primarily adopt Proximal Policy Optimization (PPO), which uses a clipped surrogate objective to ensure stable policy updates and has shown strong empirical performance in continuous control settings.

We also include Asynchronous Advantage Actor–Critic (A3C) as a secondary baseline to evaluate generalizability across architectures. However, PPO is favored for its steadier convergence and lower variance in training, especially in scenarios with highly stochastic and periodic demand. As confirmed in our experiments, PPO consistently outperforms A3C in both basic and advanced settings, justifying its selection in our final hybrid design.

Once the DRL agent selects x_t, the MSP module takes over and refines short-term decisions regarding shipment quantities z_t and vehicle allocations y_t using a scenario tree that models near-future demand uncertainty. By keeping the MSP horizon short (e.g., 4 steps), we avoid the exponential growth of scenarios and retain computational tractability. The model incorporates key operational constraints such as warehouse capacity, vehicle limits, and backorder penalties.

Crucially, the total logistics cost from the MSP step is fed back into the DRL training process as part of the reward signal. This enables the policy to learn how short-term inefficiencies propagate from earlier production decisions, and adjust accordingly over time. Through repeated interactions, the DRL agent internalizes long-term trade-offs while relying on MSP to ensure feasibility and responsiveness in short-term execution. This iterative loop allows the hybrid approach to converge to cost-effective strategies under complex, uncertain, and seasonal supply chain dynamics.

4. Experiments

In designing our experimental framework, we follow the approach of (Kemmer et al., 2018) to simulate seasonal demand within a two-echelon supply chain, consistent with prior reinforcement learning studies for supply chain planning (Chaharsooghi et al., 2008; Gijsbrechts et al., 2022). Specifically, we define the demand $d_{j,t}$ at warehouse j for timestep t as:

$$d_{j,t} = \left\lfloor D_j \left(1 + \sin\left(\frac{2\pi(t - \phi_j)}{P_j}\right)\right)\right\rfloor + \epsilon_j, \qquad \forall j = 1, \dots, J, \quad \forall t = 1, \dots, T \tag{13}$$

where $\lfloor\cdot\rfloor$ denotes the floor function, D_j is the amplitude of the baseline demand level at warehouse j (set larger than the maximum production capacity X_{max} to reflect realistic constraints (Khouja, 2003)), ϕ_j represents the phase shift, P_j indicates the period length of demand fluctuations (configured as a fraction of the planning horizon T), and ϵ_j denotes stochastic demand noise whose specific distribution is defined separately for each experimental scenario.

In our experiments, we intentionally set $\phi_j = 0$ and P_j equal for all warehouses to create the most challenging environment, characterized by synchronized peak demands. Such synchronized cycles represent a critical yet realistic case for industries like stationery manufacturing (aligned with academic calendars) or seasonal food production (Peng et al., 2019), where proactive scheduling is crucial. We assume that the stochastic terms ϵ_j are i.i.d. across all warehouses, allowing us to drop the subscript j in subsequent notation. Additionally, we ensure that strategic production planning can theoretically meet all demands within capacity constraints by enforcing:

$$\frac{P \cdot X_{max}}{J} \geq \sum_{t=0}^{P} \left\lfloor D\left(1 + \sin\left(\frac{2\pi\left(\frac{t}{P} - \phi\right)}{1}\right)\right)\right\rfloor + \bar{\epsilon}, \tag{14}$$

a feasibility condition for practical solutions.

To evaluate performance, we compare our proposed DRLBD (Deep Reinforcement Learning with Batch Decision) approach against the following baselines (Hubbs et al., 2020; Gijsbrechts et al., 2022):

- **(s, Q) Policy:** A classical inventory replenishment strategy that reorders whenever stock falls below a threshold (Zipkin, 2000).
- **Proximal Policy Optimization (PPO):** A policy-gradient RL algorithm used to optimize replenishment and shipment decisions (Fujimoto et al., 2018).
- **Asynchronous Advantage Actor–Critic (A3C):** An RL algorithm that asynchronously updates policies and value functions (Chaharsooghi et al., 2008).
- **Expected Value Problem (EVP):** Assumes perfect demand forecasting and solves a deterministic version of the SCIM problem (Alonso-Ayuso et al., 2003).
- **Perfect Information (PI) Model:** Represents an upper bound where future demand is assumed to be completely known.

4.1. Experiment Setup

To evaluate the proposed approach under different levels of complexity and realism, we conducted two types of experiments: basic scenarios and advanced scenarios, each designed to assess distinct aspects of the supply chain decision-making process.

The basic scenario involves a simplified setting with two distribution warehouses ($J = 2$) and a short planning horizon of seven timesteps, where each timestep represents one day. Demand is characterized by parameters $D = 5$ and $P = 5$, and noise is introduced through a Bernoulli distribution $\epsilon \sim \mathcal{B}(0.5)$. This controlled configuration allows us to compute the exact optimal solution using a multi-stage stochastic programming (MSP) formulation. As the scenario tree remains tractable (with $4^T = 256$ scenarios), we employ commercial solvers such as Gurobi to benchmark the performance of our heuristic against the true optimum (Preusser et al., 2010). This setting is primarily used to evaluate the optimality gap and verify the effectiveness of our method in comparison with theoretical upper and lower bounds.

The advanced scenarios simulate more realistic and challenging supply chain environments. Specifically, we consider networks with five or ten distribution warehouses ($J = 5$ or $J = 10$) over a one-year planning horizon consisting of twelve monthly timesteps ($T = 12$). Demand follows a seasonal structure defined by $D = 2$ and $P = 6$, while the noise component ϵ is sampled from a negative binomial distribution with parameters $r = 3$ and $p = 0.7$, capturing variability typically observed in practical settings (Peng et al., 2019). Due to the combinatorial explosion of scenarios, exact MSP solutions are no longer computationally feasible in this setting. Instead, we evaluate algorithm performance by comparing to a perfect-information (PI) baseline, enabling the use of the EVPI-gap as an upper-bound reference. This configuration tests the scalability, generalizability, and robustness of each method under uncertainty and structural complexity.

4.2. Experiment Results

Table 3 shows average results for the basic scenario under Bernoulli-distributed noise, focusing on the optimality gap (opt-gap) relative to the MSP solution. Our proposed DRLBD method achieves the lowest average gap of 8.25%, maintaining competitive performance with low variance. In comparison, PPO and A3C yield higher gaps of 26.91% and 30.43%, respectively, indicating their reduced robustness in the face of demand uncertainty. The (s, Q) policy and EVP perform significantly worse, with opt-gaps exceeding 66% and 120%, respectively, reinforcing the limitations of static heuristics and naive expectation-based models under stochastic conditions (Vincent, 1985).

Table 3. Average opt-gap (as a percentage) with $\epsilon \sim \mathcal{B}(0.5)$. Standard deviation in parentheses.

Algorithm	Opt-gap [%]
DRLBD	8.25 (2.4)
PPO	26.91 (10.2)
A3C	30.43 (12.3)
sQ	66.87 (31.5)
EVP	121.58 (63.2)
PI	-6.52 (3.1)

Figure 4 illustrates the distribution of optimality gaps for DRLBD and PPO in the basic scenario with Bernoulli-distributed noise. The histogram clearly demonstrates that DRLBD consistently achieves significantly lower optimality gaps compared to PPO. While the DRLBD distribution is concentrated primarily between 5-15% above optimal, the PPO distribution is centered around 30-40% above optimal, with some cases extending beyond 60%. This visualization provides compelling evidence of not only the superior average performance of DRLBD but also its greater reliability and consistency across multiple problem instances.

Figure 5 presents a comprehensive box plot comparison of cumulative costs across all tested algorithms, while Table 4 provides the precise numerical values of these costs (average of 145, 180, and 199 for DRLBD, PPO, and sQ respectively, with MS at 134 and EVP at 402). Together, they offer compelling evidence for the superiority of our hybrid approach. The plot clearly demonstrates the superior performance and consistency of both our proposed DRL-MS approach and the theoretical MS benchmark, with DRL-MS showing slightly tighter variance bounds. The standalone DRL approach shows moderately higher costs with slightly wider dispersion, while the sQ policy exhibits both higher median costs and greater variability. As expected, the EVP method displays the poorest performance with extremely high variability and numerous outliers, confirming the critical importance of explicitly modeling uncertainty in supply chain decision-making. Meanwhile, the Perfect Information (PI) model, which assumes complete knowledge of future demand, establishes a lower bound that is theoretically unattainable in practice but serves as a useful reference point.

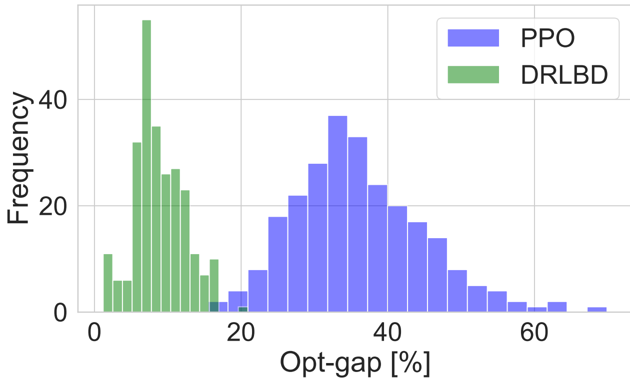

Figure 4. Histogram showing the frequency distribution of optimality gaps for DRLBD and PPO over 250 episodes in the basic scenario with Bernoulli-distributed noise. X-axis shows optimality gap percentage (0-60%), Y-axis shows frequency.

Table 4. Average cumulative costs for all algorithms across 250 episodes. Standard deviation in parentheses.

Algorithm	Cumulative Costs
DRLBD	145 (5)
PPO	180 (13)
sQ	199 (24)
MS	134 (5)
EVP	402 (222)
PI	89 (12)

In the advanced scenario, Table 5 presents the expected value of perfect information gap (EVPI-gap) relative to the PI benchmark. DRLBD continues to outperform all baselines with the smallest gaps (102.46% for $J = 5$, 111.37% for $J = 10$), while other methods—particularly EVP and sQ—suffer from pronounced degradation as the network scales. Although DRLBD's performance slightly declines compared to the basic scenario, it remains significantly more robust than standalone DRL approaches like PPO and

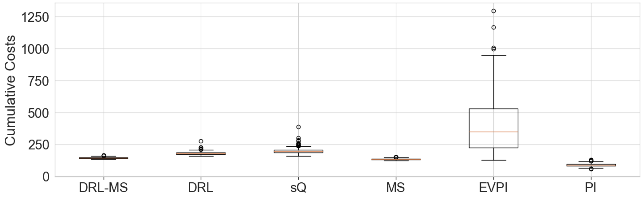

Figure 5. Box plot showing the distribution of cumulative costs for different algorithms (DRL-MS, DRL, sQ, MS, EVPI, PI). The y-axis represents cumulative costs, with lower values indicating better performance.

A3C, whose gaps increase more steeply with J. These results emphasize the resilience and scalability of the hybrid DRL-MSP method in managing uncertainty across complex supply chain configurations (Preusser et al., 2010).

Table 5. Average EVPI-gap (%) for advanced scenarios over 250 episodes. Standard deviation in parentheses.

Algorithm	**EVPI-gap (%)** $J = 5$	**EVPI-gap (%)** $J = 10$
DRLBD	102.46 (23)	111.37 (20)
PPO	137.52 (35)	146.18 (28)
A3C	141.23 (33)	150.40 (29)
sQ	151.90 (36)	194.05 (44)
EVP	555.67 (166)	584.92 (130)

Figure 6 presents the total cost distributions for DRLBD and PPO in the advanced scenario with J = 5 warehouses. The histogram reveals a clear separation between the two approaches, with DRLBD achieving consistently lower costs centered around 140-150 thousand euros, while PPO costs are primarily distributed between 170-190 thousand euros. This substantial cost difference demonstrates the practical economic impact of our hybrid approach in realistic supply chain settings.

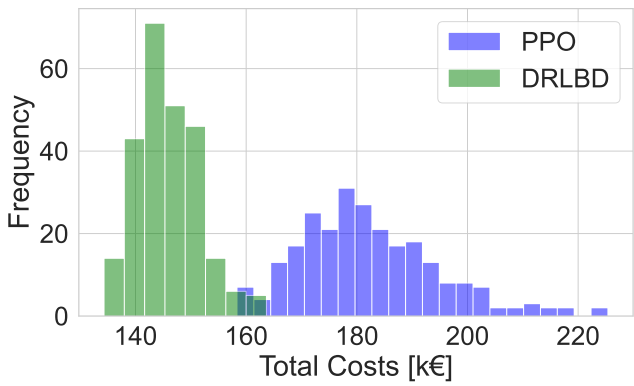

Figure 6. Histogram showing the frequency distribution of total costs for DRLBD and PPO over 250 episodes in the advanced scenario with J = 5. X-axis shows total cost (in thousands of euros), Y-axis shows frequency.

Table 6 provides a comparison of average training and testing times for the advanced scenario with $J = 10$. As expected, DRL-based approaches incur higher training costs; however, once trained, they require only minimal time per episode to make decisions (Hubbs et al., 2020). Notably, the Perfect Information model (PI) shows high variance in testing times due to solving the full deterministic optimization repeatedly.

Figure 7 illustrates the training convergence of the PPO algorithm in terms of mean reward per episode. The reward is defined as the negative of total cost, so higher (less negative) values indicate better performance. As shown in the figure, the algorithm exhibits rapid improvement during the first 20,000 episodes, followed by continued but more gradual enhancements until approximately 38,000 episodes, after which performance stabilizes. This convergence pattern is consistent with the training dynamics typically observed in deep reinforcement learning applications.

Table 6. Average training (min) and testing (ms) times for $J = 10$. Standard deviation in parentheses.

Algorithm	**Train [min]**	**Test [ms]**
DRLBD	7.15 (0.02)	33 (2.10)
PPO	7.10 (0.02)	24 (1.46)
A3C	7.85 (0.03)	29 (1.82)
sQ	6.40 (0.01)	13 (0.62)
EVP	–	58 (0.74)
PI	–	310 (645)

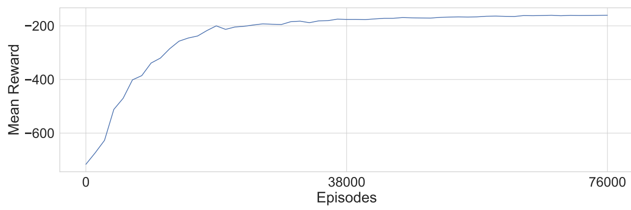

Figure 7. Line graph showing mean reward versus training episodes for PPO, with x-axis "Episodes" ranging from 0 to 76000 and y-axis "Mean Reward" showing values from approximately -700 to -200.

4.3. Discussion

Overall, the experimental results consistently demonstrate that DRLBD outperforms classical heuristics and standalone DRL baselines across both basic and advanced settings. In the basic scenario, where exact MSP solutions are available, DRLBD achieves the lowest average optimality gap with narrow standard deviation margins, indicating not only superior performance but also high consistency. In the more complex advanced scenario, DRLBD maintains the lowest EVPI-gap under both $J = 5$ and $J = 10$ conditions, further reinforcing its robustness as problem scale increases.

Importantly, all results are averaged over 250 episodes to ensure statistical significance, and standard deviations reported in Tables 3 and 5 confirm the stability of our findings. For instance, DRLBD exhibits a standard deviation of only 2% in the basic scenario and less than 20 in EVPI-gap across advanced cases, which is markedly lower than that of other baselines. These differences suggest not only better average performance, but also more reliable decision quality over repeated simulations.

By integrating long-horizon decisions from DRL with short-term adjustments via multi-stage stochastic programming (Alonso-Ayuso et al., 2003; Stranieri et al., 2024), the proposed hybrid method effectively balances adaptability and precision. The results further highlight DRLBD's ability to exploit seasonality and cope with demand uncertainty without requiring exhaustive scenario enumeration, making it well-suited for real-world deployment where scalability and interpretability are essential.

5. Related Work

Supply Chain Inventory Management (SCIM) is a classical yet challenging problem, driven by sequential decision-making and stochastic demand. Several solution paradigms have been proposed, most notably multi-stage stochastic programming (MSP), heuristic reordering rules, and reinforcement learning (RL).

MSP models capture demand uncertainty via scenario trees and provide optimal solutions under certain conditions (Khouja, 2003; Huang et al., 2010). However, they suffer from the curse of dimensionality, particularly when modeling long horizons or seasonal patterns (Alonso-Ayuso et al., 2003), making them less scalable for realistic applications.

Heuristic approaches such as the (s, S) and (s, Q) policies remain popular due to their simplicity and interpretability (Zipkin, 2000). Nevertheless, these static rules are limited in their ability to respond to dynamic environments with high variability or evolving constraints.

In recent years, reinforcement learning has emerged as a data-driven alternative to traditional methods. Deep reinforcement learning (DRL) techniques, such as Proximal Policy Optimization (PPO) (Hubbs et al., 2020) and Asynchronous Advantage Actor-Critic (A3C) (Gijsbrechts et al., 2022), have shown promise in complex supply chain environments. However, DRL methods are often sensitive to hyperparameters and may struggle to incorporate constraints like production capacity or logistics cost structures (Fujimoto et al., 2018).

To address these limitations, hybrid methods have been explored. For instance, Bertsekas (Bertsekas, 2021) proposed using approximate dynamic programming alongside stochastic optimization, while Harsha et al. (Harsha et al., 2021) integrated RL with mixed-integer linear programming to manage constrained inventory systems.

Still, most existing approaches fail to capture all key SCIM features, such as seasonal demand, production bottlenecks, and storage trade-offs. Notably, Peng et al. (Peng et al., 2019) applied DRL in a seasonal two-echelon supply chain but omitted logistics and capacity constraints.

Our work builds on this foundation by proposing a DRL-MSP hybrid that combines the adaptability of DRL for long-term production with the structure of MSP for short-term logistics. The resulting framework is designed to handle uncertainty, non-stationarity, and operational constraints in a unified manner.

6. Conclusion

This report proposes a hybrid framework for supply chain inventory management in a two-echelon divergent network, combining deep reinforcement learning (DRL) for long-term production planning with multi-stage stochastic programming (MSP) for short-term logistics optimization. By leveraging the adaptability of DRL and the robustness of MSP, the method effectively addresses uncertainty in dynamic supply chains.

Experimental results across both small-scale and large-scale settings demonstrate that the proposed DRL-MSP approach (DRLBD) consistently outperforms benchmark methods, including PPO, A3C, and (s, Q) policies. In scenarios where the optimal solution is known, DRLBD achieves near-optimal performance; in more complex environments, it maintains a significantly lower EVPI-gap.

While PPO slightly outperforms A3C in some cases, both are surpassed by DRLBD, particularly as problem scale and uncertainty increase. Although DRL-based methods involve higher training costs, their fast inference enables efficient real-time decision-making and what-if analysis.

6.1. Limitations and Future Work

The DRLBD framework demonstrates strong performance in managing supply chain inventory under uncertainty. However, several limitations suggest directions for future research. The current model does not consider fixed production setup costs, stochastic lead times, or multi-echelon supply chain structures. Extending the framework to support more complex network topologies, such as intermediate hubs or multi-product flows, would enhance its practical relevance. In addition, logistics parameters are assumed to be known and static.

Future work could integrate learning-based methods to dynamically estimate transportation costs, lead times, or capacity constraints, improving adaptability in real-time operations. Enhancing the robustness and generalization of the DRL component is also a promising direction. Techniques like meta-reinforcement learning or curriculum learning could help adapt to evolving demand patterns and partial observability. Finally, incorporating factors such as emissions constraints, fixed production costs, and supplier reliability would further increase the model's realism and industrial applicability.

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